The University of Azad Jammu and Kashmir

Department of Software Engineering

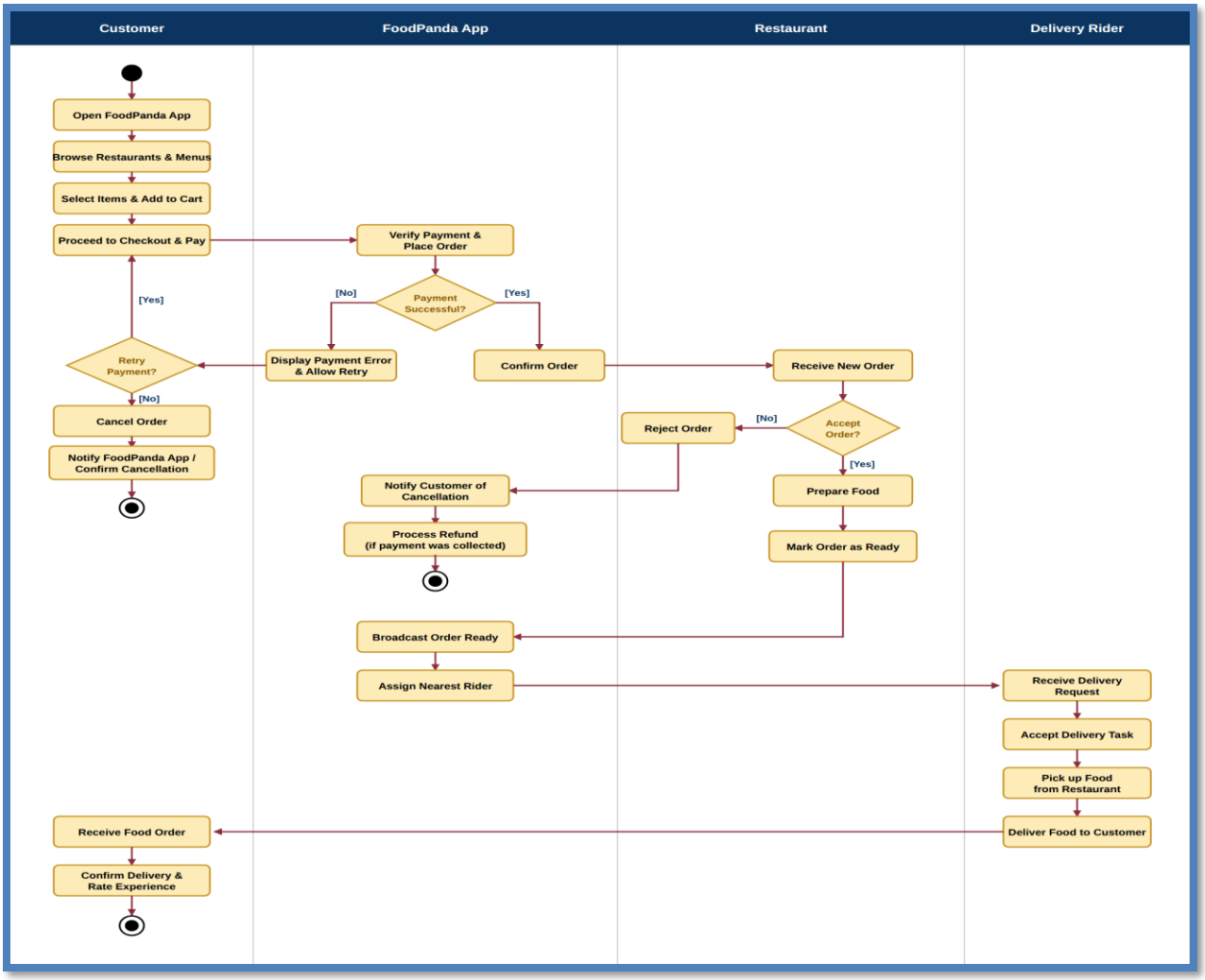
Student Name	Afzal Ali Ahmed Shah
Roll no	2024-SE-04
Semester	5th
Instructor Name	Engr. Asma Javaid
Subject	Software Construction and Development
Course Code	SE-3101
Credit Hours	2+1
Lab	02
Date	25-Oct-2026

Department of Software Engineering

Question 1: FoodPanda Online Food Order App (Activity Diagram)

Task Requirement: Draw an activity diagram by visualizing the process of the FoodPanda application (online food order app) using swimlanes to define actor responsibilities across Customer, FoodPanda App, Restaurant, and Delivery Rider.

Swimlane / Partition	Key Operational Responsibilities
Customer	Open app, browse restaurants/menus, select items and add to cart, proceed to checkout and pay, decide whether to retry payment after an error (cancel order and notify the app if not retrying), receive food order, and confirm delivery/rate experience.
FoodPanda App	Verify payment and place order, display payment error and allow retry, confirm order and forward it to the restaurant, notify customer of cancellation and process refund (if payment was collected) when the restaurant rejects, broadcast order ready status, and assign nearest rider.
Restaurant	Receive new order, decide whether to accept the order, reject the order if not accepted, prepare food upon acceptance, and mark order as ready.
Delivery Rider	Receive delivery request, accept delivery task, pick up food from restaurant, and deliver food to customer.

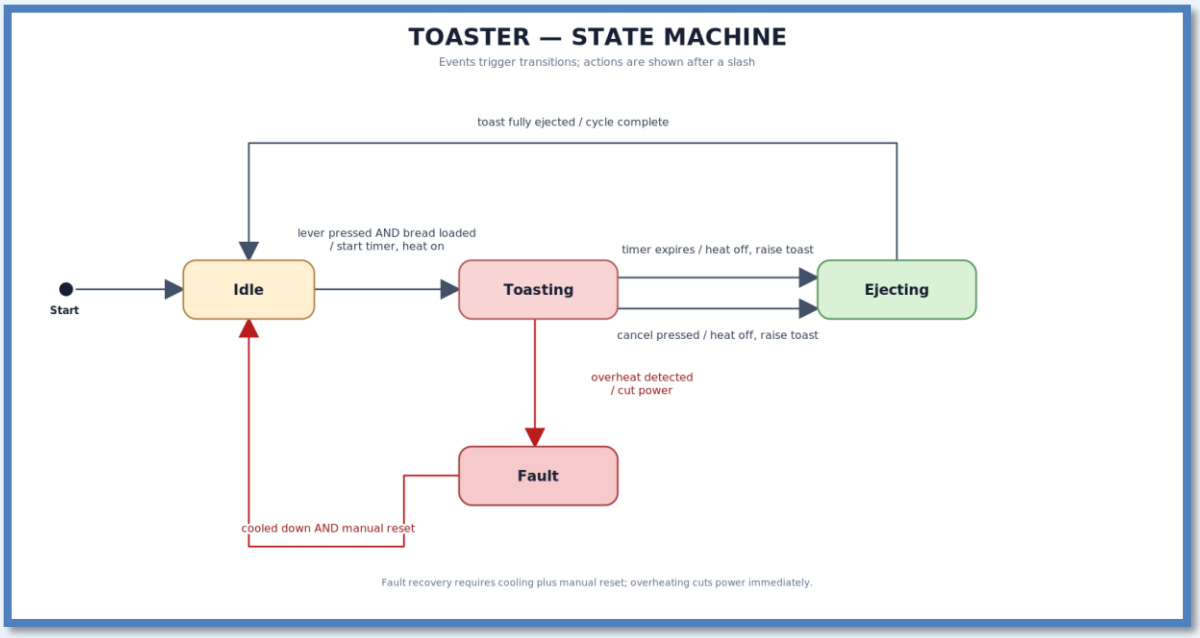


Question 2: Toaster State Machine Diagram

Task Requirement: Draw a State Machine diagram for a Toaster. List activities within each state (entry, do, exit) and explicitly identify triggered versus triggerless completion events/actions.

State Name	Internal Activities (entry / do / exit)	Outgoing Transitions & Triggers
Idle	entry / Keep heating element off do / Wait for bread and lever press	--> Toasting : Lever pressed AND bread loaded / start timer, heat on (Triggered)
Toasting	entry / Turn heating element on and start timer do / Heat bread and monitor timer and temperature exit / Turn heating element off	--> Ejecting : Timer expires / heat off, raise toast (Triggered) --> Ejecting : Cancel pressed / heat off, raise toast (Triggered)

		--> Fault : Overheat detected / cut power (Triggered)
Ejecting	entry / Release latch do / Raise bread carriage and eject toast exit / Reset timer	--> Idle : Toast fully ejected / cycle complete (Triggered)
Fault	entry / Cut power to heating element do / Stay locked out until toaster cools down	--> Idle : Cooled down AND manual reset (Triggered)



Question 3: Elevator/Lift State Machine Diagram

Task Requirement: Draw a State Machine diagram of an Elevator/Lift system representing internal behaviors, directional travel states, door security cycles, and user call triggers.

State Name	Internal Behaviors (entry / do / exit)	Transition Triggers & Target States
Idle	entry / Ensure car is stopped and doors are closed do / Wait for floor request	--> Moving Up : Request above (Triggered) --> Moving Down : Request below (Triggered)

		--> Doors Open : Request at current floor (Triggered)
Moving Up	entry / Start motor upwards do / Update floor indicator	--> Doors Open : Arrived at target floor (Triggered) --> Emergency Stop : Emergency button (Triggered)
Moving Down	entry / Start motor downwards do / Update floor indicator	--> Doors Open : Arrived at target floor (Triggered) --> Emergency Stop : Emergency button (Triggered)
Doors Open	entry / Unlock and open doors, start timer do / Wait for passengers exit / Close doors	--> Idle : Timer elapsed (Triggered)
Emergency Stop	entry / Stop motor, apply brake and sound alarm do / Hold car stopped	--> Idle : Emergency cleared AND technician reset (Triggered)

